

Countable Mixed Dual Sums Satisfy the Sobolev Weak* Differentiability Conclusion

Candidate strong partial result — `fa_banach_001`, lane 3

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Status. Strong partial result, likely valid. The general source question remains open. Human review is recommended.

1 Source question

Creutz and Evseev study Sobolev maps into a dual Banach space V^* [1]. They define the *weak* Lebesgue density property* and prove that it is equivalent to almost-everywhere weak* differentiability of every absolutely continuous curve into V^* . At the end of page 12 they ask whether this one-dimensional property implies that every $f \in W^{1,p}(\Omega; V^*)$ has a representative which is weak* differentiable almost everywhere in the coordinate directions.

whenever the latter exists. Then $\nu \equiv_* \eta$ and for every t at which γ is weak* differentiable

$$\int_t^{t+h} \nu(s) \, ds = \int_t^{t+h} \eta(s) \, ds = \frac{1}{h} \cdot (\gamma(t+h) - \gamma(t)) \xrightarrow{*} \nu(t)$$

as $h \rightarrow 0$. □

If V^* is dual-to-separable or has the Radon–Nikodým property then (as was previously observed in [20, 5, 10] resp. [6, 12]) it follows that maps $f \in W^{1,p}(\Omega; V^*)$ have representatives that are weak* differentiable almost everywhere. It remains an open question whether this is implied more generally by the weak* Lebesgue density property. As in Section 3.4 the problem is that a nonmeasurable property on Ω may hold almost everywhere along almost every line segment.

REFERENCES

- [1] Stephanie Alexander, Vitali Kapovitch, and Anton Petrunin. Alexandrov geometry: foundations. Preprint arXiv:1903.08539v5, 2009.

Figure 1: The open question in arXiv:2303.17303v1, page 12.

Their two known positive cases are: the predual V is separable, or the dual V^* has the Radon–Nikodým property (RNP). The theorem below closes the question for every countable direct sum made by mixing these two kinds of summands.

2 Statement and proof intuition

Let $(V_k)_{k \in \mathbb{N}}$ be Banach spaces and put $X_k = V_k^*$. For $1 \leq s \leq \infty$ set

$$X_s = \left(\bigoplus_{k \in \mathbb{N}} X_k \right)_{\ell^s}.$$

We always use the canonical predual

$$E_s = \begin{cases} (\bigoplus_k V_k)_{c_0}, & s = 1, \\ (\bigoplus_k V_k)_{\ell^{s'}}, & 1 < s \leq \infty, \end{cases} \quad X_s = E_s^*,$$

where s' is conjugate to s and $s' = 1$ when $s = \infty$.

Proof intuition. The source obstruction is measurability: weak* differentiability tests all elements of a potentially nonseparable predual, so linewise full-measure sets need not combine into a measurable full-measure subset of Ω . A countable mixed sum has an extra coordinate structure. In a separable-predual coordinate, countably many scalar tests suffice. In an RNP coordinate, norm differentiability is itself a measurable property. There are only countably many coordinates, so their good sets have a measurable full-measure intersection. At such a point, the coordinate derivatives are assembled by finite-coordinate weak* lower semicontinuity. Approximation by finite-support predual vectors then gives weak* convergence in the whole sum. This proves a substantial new class but does not resolve the arbitrary-space measurability obstruction.

Theorem 1 (countable mixed-sum theorem). *Assume that for every $k \in \mathbb{N}$ at least one of the following holds:*

1. V_k is separable;
2. $X_k = V_k^*$ has the Radon–Nikodým property.

Then, for every $1 \leq s \leq \infty$, the canonical dual $X_s = E_s^$ has the weak* Lebesgue density property. Moreover, if $\Omega \subset \mathbb{R}^n$ is a bounded domain, $1 \leq p < \infty$, and $f \in W^{1,p}(\Omega; X_s)$, then f has a representative which is weak* differentiable almost everywhere in each coordinate direction.*

We separate the assembly argument used twice in the proof.

Lemma 2 (coordinate assembly). *Let $q_h \in X_s$ for $0 < |h| < h_0$, assume $\sup_{0 < |h| < h_0} \|q_h\| \leq M$, and suppose that for every k there is $d_k \in X_k$ such that $\pi_k q_h \xrightarrow{w^*} d_k$ as $h \rightarrow 0$. Then $d = (d_k)_k \in X_s$, $\|d\| \leq M$, and $q_h \xrightarrow{w^*} d$ in the duality $X_s = E_s^*$.*

Proof. For a finite set $F \subset \mathbb{N}$, coordinatewise convergence is weak* convergence in the finite dual sum. Weak* lower semicontinuity of its norm gives

$$\|(d_k)_{k \in F}\|_{\ell^s} \leq \liminf_{h \rightarrow 0} \|(\pi_k q_h)_{k \in F}\|_{\ell^s} \leq M.$$

Taking the supremum over finite F shows $d \in X_s$ and $\|d\| \leq M$; this includes $s = \infty$.

Let $v \in E_s$ and let v^F be its truncation to a finite initial coordinate set F . Finite-coordinate convergence gives $\langle v^F, q_h - d \rangle \rightarrow 0$, while

$$|\langle v - v^F, q_h - d \rangle| \leq \|v - v^F\|_{E_s} (\|q_h\| + \|d\|).$$

Finite sequences are dense in E_s : this is the usual fact for $\ell^{s'}$ when $1 < s \leq \infty$, and for the c_0 -sum when $s = 1$. Letting $F \uparrow \mathbb{N}$ proves the full weak* convergence. $\square$

Proof of Theorem 1. We first prove the weak* Lebesgue density property. By the source theorem [1, final theorem of Section 3.5], it suffices to show that every absolutely continuous curve $\gamma : (a, b) \rightarrow X_s$ is weak* differentiable almost everywhere. Each coordinate curve $\gamma_k = \pi_k \gamma$ is absolutely continuous. If V_k is separable, it is weak* differentiable almost everywhere by a countable dense family of tests in V_k . If X_k has the RNP, it is norm differentiable almost everywhere. Intersect these full-measure sets over the countably many k , and also intersect with the set where the metric derivative of γ exists. At a resulting point t the difference quotients

$$q_h = \frac{\gamma(t+h) - \gamma(t)}{h}$$

are bounded and converge weak* in every coordinate. Lemma 2 gives a weak* derivative in X_s . The cited equivalence now yields the weak* Lebesgue density property.

We turn to Sobolev maps. Choose the standard measurable ACL representative $\tilde{f}$ supplied by the metric-valued Sobolev ACL theorem used in [1, Proposition 2.4]. Fix a coordinate direction e_j . The metric partial derivative

$$\text{m } \partial_j \tilde{f}(x) = \lim_{h \rightarrow 0} \frac{\|\tilde{f}(x + he_j) - \tilde{f}(x)\|}{|h|}$$

exists at almost every x and is measurable.

We construct a measurable full-measure set for each target coordinate k . If V_k is separable, choose a countable dense rational linear subspace $D_k \subset V_k$. For every $v \in D_k$, the scalar ACL function $x \mapsto \langle v, \pi_k \tilde{f}(x) \rangle$ has its j -th partial derivative almost everywhere. Intersecting these good sets over D_k and with the metric-partial-derivative set gives a measurable full-measure set. At one of its points the X_k -valued difference quotients are bounded, and their scalar limits on D_k define a bounded functional on D_k which extends uniquely to V_k . Density then gives weak* convergence in X_k .

If X_k has the RNP, the coordinate map $\pi_k \tilde{f}$ is absolutely continuous on almost every e_j -parallel line and is norm differentiable almost everywhere on each such line. The corresponding set in Ω is measurable: on an ACL line, existence of the norm derivative is the norm-Cauchy condition for difference quotients with rational increments, a countable measurable condition. Fubini therefore produces a measurable full-measure set on which the k -th coordinate difference quotient converges in norm.

Intersect these measurable sets over all k and with the metric-partial- derivative set. This is still a measurable full-measure subset of Ω . At any point x in it, put

$$q_h(x) = \frac{\tilde{f}(x + he_j) - \tilde{f}(x)}{h}.$$

The quotients are bounded for small h and converge weak* in every target coordinate. Lemma 2 gives a weak* j -th partial derivative of $\tilde{f}$ at x . Finally intersect over the finitely many domain directions $j = 1, \dots, n$. $\square$

3 A space beyond both source hypotheses

Corollary 3. *Let Γ have cardinality strictly larger than the continuum and set*

$$X = \ell^\infty(\mathbb{N}) \oplus_2 \ell^1(\Gamma) = (\ell^1(\mathbb{N}) \oplus_2 c_0(\Gamma))^*.$$

Then X has the weak Lebesgue density property and satisfies the Sobolev weak* differentiability conclusion of Theorem 1, but X has no RNP and is not a dual of any separable Banach space.*

Proof. The first summand is the dual of the separable space $\ell^1(\mathbb{N})$, while the second has the RNP. Theorem 1 applies. The complemented subspace $\ell^\infty(\mathbb{N})$ prevents X from having the RNP. Finally, a dual of a separable Banach space has cardinality at most the continuum, whereas X contains the family $(e_\gamma)_{\gamma \in \Gamma}$. Hence X admits no separable predual. $\square$

4 Scope, verification, and novelty

The theorem does not prove that an arbitrary weak* Lebesgue-density dual has the desired Sobolev differentiability property. It also does not treat uncountable direct sums. Its contribution is a countable stability mechanism which yields explicit spaces outside both sufficient classes listed in the source.

The proof is exact and uses no numerical computation. The endpoint preduals c_0 and ℓ^1 are handled explicitly in Lemma 2. The source question and the later related preprint [2] were searched by title, arXiv id, exact question phrase, and combinations of “weak* Lebesgue density”, “direct sum”, and “Sobolev weak* differentiability”. The local run indexes were searched by the same terms. No later resolution or prior statement of this mixed-sum theorem was found. This is a bounded novelty check, not exhaustive bibliographic certification.

Human-review recommendation. Verify especially the measurability of the RNP-coordinate norm-differentiability set for the common ACL representative, and the finite-support tail argument at $s = 1$ and $s = \infty$. These are isolated in the proof so that they can be audited without reconstructing the discovery process.

References

- [1] P. Creutz and N. Evseev, *Weak differentiability of metric space valued Sobolev maps*, arXiv:2303.17303v1, 2023.
- [2] N. Evseev, *Weak derivatives and metric differentiability almost everywhere*, arXiv:2511.02520v1, 2025.